

Title: Task 4 – Local Deployment of Containerized Application using Docker and WSL 2.

Application Name: PakMilk (Fresh Dairy Online Platform).

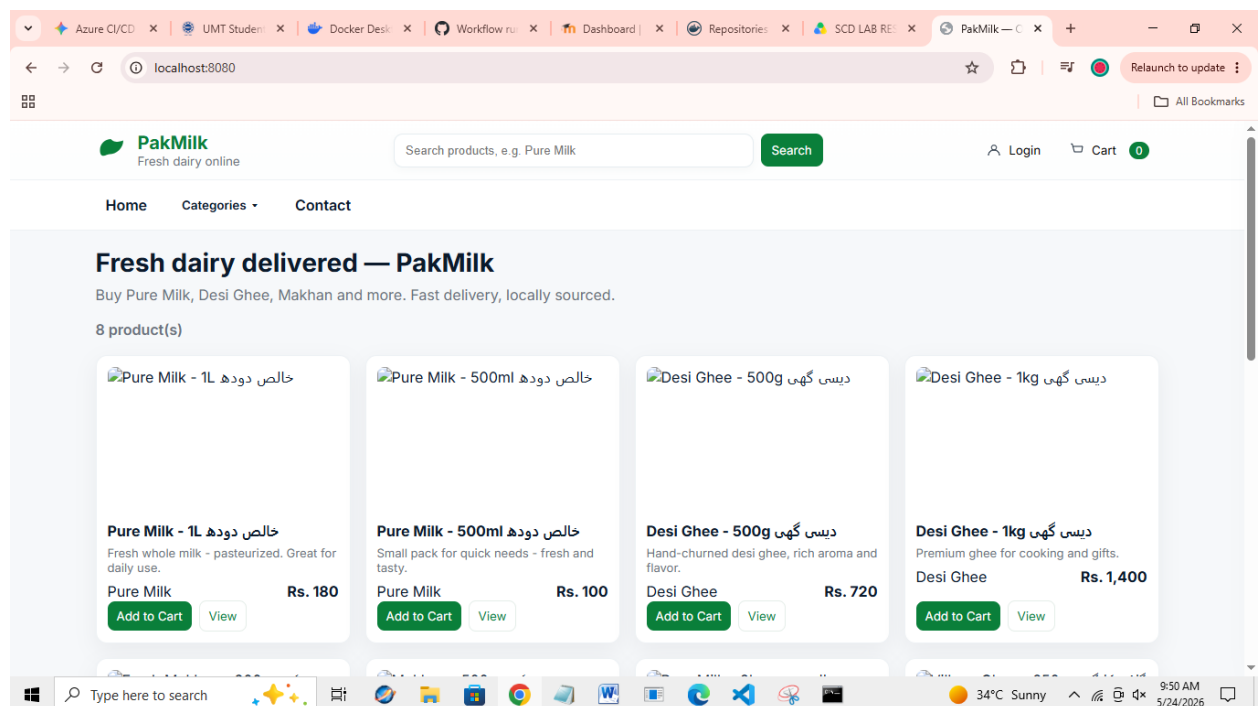
Objective: To pull the built Docker image from Docker Hub and deploy it locally inside an isolated Linux environment using Windows Subsystem for Linux (WSL 2) and Docker Engine, ensuring the web application serves traffic successfully on port 8080.

Technical Steps Executed (Briefly list these)

- Initialized **Ubuntu 26.04 LTS** via WSL 2 on Windows.
- Installed the **Docker Engine** utilities directly inside the Linux environment using `apt-get`.
- Started the background Docker service daemon (`sudo service docker start`).
- Pulled and ran the production-ready image from Docker Hub using:

```
sudo docker run -d -p 8080:80 --name pakmilk_server hmadrana/pakmilk
```

Resolved internal container port routing (mapping host port 8080 to container Nginx port 80) to ensure a stable 200 OK network response.



```
dell_inspiron1234@DESKTOP-HDUNCTD: /mnt/c/Users/Dell Inspiron$
Processing triggers for libc-bin (2.43-2ubuntu2) ...
dell_inspiron1234@DESKTOP-HDUNCTD: /mnt/c/Users/Dell Inspiron$ sudo service docker start
dell_inspiron1234@DESKTOP-HDUNCTD: /mnt/c/Users/Dell Inspiron$ sudo docker run -d -p 8080:8080 hmadrana/pakmilk
Unable to find image 'hmadrana/pakmilk:latest' locally
latest: Pulling from hmadrana/pakmilk
be5351d87f1f: Pull complete
53bfd6d6de95: Pull complete
22ed557e9038: Pull complete
6a0ac1617861: Pull complete
42f24dde8a4: Pull complete
19af4ef65048: Pull complete
02eda4087d06: Pull complete
00ffc65fde23: Pull complete
7f32a4497b31: Pull complete
Digest: sha256:b685f85e86438f0508bfaa3a1e4ccf7420a691f4b2346706035527928a85a5f7
Status: Downloaded newer image for hmadrana/pakmilk:latest
06305c86abdeba98fbb15557e19642e4263f99fbf65fcab7a7ed390d8949d10d
dell_inspiron1234@DESKTOP-HDUNCTD: /mnt/c/Users/Dell Inspiron$ sudo docker logs 0630
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/05/24 04:25:13 [notice] 1#1: using the "epoll" event method
2026/05/24 04:25:13 [notice] 1#1: nginx/1.31.0
2026/05/24 04:25:13 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/05/24 04:25:13 [notice] 1#1: OS: Linux 6.6.114.1-microsoft-standard-WSL2
2026/05/24 04:25:13 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/05/24 04:25:13 [notice] 1#1: start worker processes
2026/05/24 04:25:13 [notice] 1#1: start worker process 30
2026/05/24 04:25:13 [notice] 1#1: start worker process 31
2026/05/24 04:25:13 [notice] 1#1: start worker process 32
2026/05/24 04:25:13 [notice] 1#1: start worker process 33
dell_inspiron1234@DESKTOP-HDUNCTD: /mnt/c/Users/Dell Inspiron$ sudo docker stop 0630
0630
dell_inspiron1234@DESKTOP-HDUNCTD: /mnt/c/Users/Dell Inspiron$ sudo docker rm 0630
0630
dell_inspiron1234@DESKTOP-HDUNCTD: /mnt/c/Users/Dell Inspiron$ sudo docker run -d -p 8080:80 --name pakmilk_server hmadrana/pakmilk
ca14dbe7f183913d51de9cc3641d7a22b709c0c0ba2ac0506707597af95e9aaa
dell_inspiron1234@DESKTOP-HDUNCTD: /mnt/c/Users/Dell Inspiron$
```

